Experiment 1: Room Reverberation (LTI)

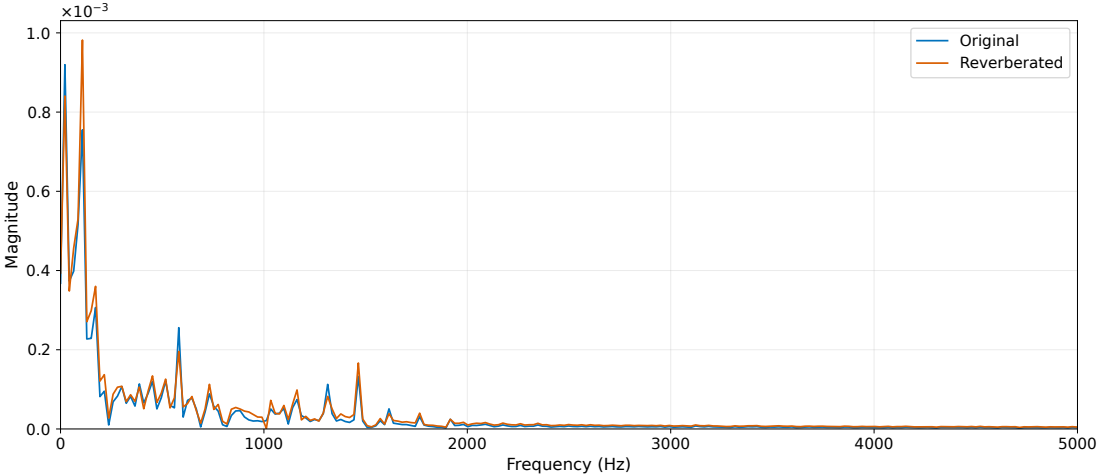
- We read the guitar recording at $f_s = 44.1$ kHz. A room adds weaker copies that arrive later than the direct sound.
- An arrival time t_i seconds gives sample delay $d_i = \text{round}(t_i f_s)$. For example, 30 ms gives 1,323 samples.
- We model the room impulse response and calculate the output by manual convolution:

$$h[n] = \delta[n] + \sum_{i=1}^{5} a_i \delta[n - d_i], \qquad y[n] = (x * h)[n] = x[n] + \sum_{i=1}^{5} a_i x[n - d_i].$$

- The gains and delays stay fixed, so the system is LTI. Scaling or adding inputs scales or adds outputs. Shifting the input shifts the output equally.
- Short delays blend the copies. Longer delays can make echoes clearer. The full output includes about 145 ms after the input ends.

Delays: 30, 55, 80, 110, 145 ms. Corresponding gains: 0.5, 0.4, 0.3, 0.2, 0.1.

Experiment 1: Magnitude-Spectrum Comparison



Original and reverberated signals: the same 2,048 samples (46.44 ms).
Direct DFT. Section: 0.07785–0.12429 s. Compare the heights of the peaks.

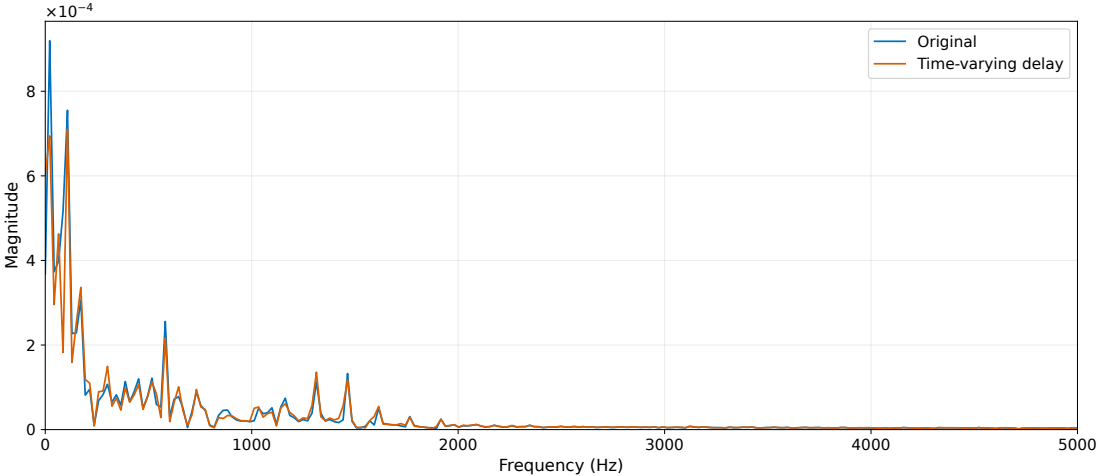
Experiment 2: Time-Varying Delay (LTV)

- We mix the original with one weaker copy whose delay changes with time:

$$D[n] = \frac{f_s}{1000}\left[D_0 + \Delta D \sin\left(\frac{2\pi f_m n}{f_s}\right)\right], \qquad y[n] = x[n] + \alpha x[n - D[n]].$$

- D_0 and ΔD are in milliseconds. $D[n]$ is in samples. The factor $f_s/1000$ converts milliseconds to samples.
- Settings: $D_0 = 15$ ms, $\Delta D = 5$ ms, $f_m = 0.5$ Hz, $\alpha = 0.5$. The delay varies from 10 to 20 ms and repeats every 2 seconds.
- We calculate the output sample by sample. If $n - D[n]$ lies between two samples, linear interpolation uses their weighted average.
- The delay depends on time, so shifting the input can change the delay it receives. The system is LTV and remains linear.
- Larger ΔD gives a wider delay range. Higher f_m makes it vary faster. The code also tries $\Delta D = 2$ ms, giving 13–17 ms delays.

Experiment 2: Magnitude-Spectrum Comparison



Original and time-varying-delay output: the same 2,048 samples (46.44 ms).
Direct DFT. Section: 0.07785–0.12429 s. $D_0 = 15$ ms, $\Delta D = 5$ ms, $f_m = 0.5$ Hz, $\alpha = 0.5$.